AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM359

Systems Penetration Testing

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Final Examination
Spring Semester 2025/2026

Date: 3 /Jun./2026

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 3 h

Instructions:

- Total Marks: 50
- Total Number of Questions:7
- Marks Distribution:
  - Part 1: 10 Marks
  - Part 2: 30 Marks
  - Part 3: 10 Marks
- Write your answer in the answer booklets. Answers given on the exam paper will not be marked.
- The use of electronic devices that could have a memory is NOT permitted.
- Be sure you write your name and ID on the External Answer booklet.

PART 1: ALL QUESTIONS ARE REQUIRED [10 Marks]

Question 1: Choose one option for each statement: True or False. [5 Marks]

A. ARP poisoning can be used to perform a man-in-the-middle attack.
B. Hash functions are reversible if the correct key is known.
C. A crypter does not protect malware from reverse engineering and analysis.
D. Multi-factor authentication reduces the risk of credential theft.
E. Denial-of-Service attacks always require malware installed on the victim system.

Question 2: Fill in each blank with one correct word, acronym, name, or phrase. Write only one answer per blank in the answer booklet. More than one answer may be correct. Each blank is worth 1 mark. [5 Marks]

A. In DNS resolution, the client typically sends a_____ request.
B. The process of gathering publicly available information about a target is called _____.
C. The protocol commonly used to securely access a remote Linux server is_____.
D. The principle of granting only the minimum required permissions is called_____.
E. A digital certificate is commonly issued by a _____.

PART 2: ALL QUESTIONS ARE REQUIRED [30 Marks]

Question 1: Answer the following questions.

Question 1 (a): Analyze how a Trojan infects a system using the structured SIX (6) steps. [6 Marks]

Question 1 (b): List down FOUR (4) characteristics of Advanced Persistent Threats (APTs). Your answer should address the objective, timeliness, phase, and point of entry involved in the APT attack. [4 Marks]

Question No. 2: Analyze Figure 1 as follows, and answer the following questions:

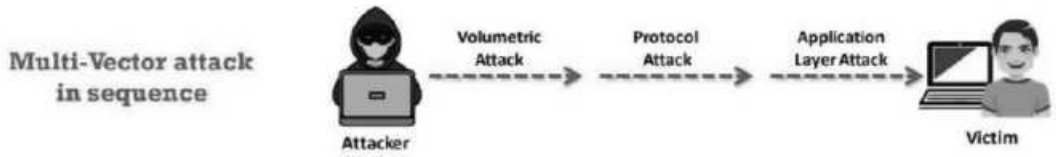


Figure 1: Multi-Vector Attack

Question No. 2 (a): Based on the Figure 1, analyze how different DDoS attack vectors target various layers of the victim's infrastructure. Briefly explain each attack vector and provide objective of each attack vector. [6 marks]

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Question 2. (b): Explain how combining volumetric, protocol, and application-layer attacks in a multi-vector DDoS attack increases the overall impact on the victim's infrastructure. [4 Marks]

Question 3: Answer the following:

Question 3. (a). Define social engineering and identify two common targets of social engineering attacks. [3 Marks]

Question 3. (b): Explain the FOUR (4) phases of a social engineering attack. [4 Marks]

Question 3. (c): Describe any THREE (3) social engineering countermeasures that organizations can adopt to prevent social engineering attacks.[3 Marks]

PART 3: ALL QUESTIONS ARE REQUIRED [10 Marks]

Question 1 (a): Define encryption in Cryptography. [2 Marks]

Question 1 (b): Differentiate between Infrastructure as a Service (IaaS) and Platform as a Service (PaaS) in cloud computing. [3 Marks]

Question 2: Match each term in Column A with the correct description in Column B. Write the correct letter and number (For Example: A1 → B3). Each correct match is worth 1 mark. [5 Marks]

Column A	Column B
A1. Brute-force attack	B1. A network security device that filters traffic based on rules
A2. Firewall	B2. A collection of compromised machines controlled remotely
A3. Zero-day vulnerability	B3. A vulnerability unknown to the vendor at the time of exploitation
A4. Botnet	B4. A method of trying many password combinations systematically
A5. Digital Signature	B5. Provides integrity and non-repudiation using cryptographic keys

End of Question Paper

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